Does a state clamp stabilise the free net?

exp064 · 11 July 2026 · Draft

Abstract

The free signed-recurrent network diverges during training: its loss becomes a not-a-number (NaN). Reading the forward pass locates the cause. The conductance-based neuron relaxes each timestep toward a steady-state voltage v_∞ that is inversely proportional to the total conductance g_{tot}, so it blows up when signed weights drive g_{tot} to zero or below. The fault is the **sign** of the weights, not their size. The hypothesis: a forward-pass state clamp that floors every conductance at 0 each timestep keeps $g_{\text{tot}} \geq g_L > 0$ and removes the divergence, while the weights stay signed and expressive. Three cells test it at $\Delta t = 1$ ms, single seed: the free baseline, free plus clamp, and clamp plus strong weight decay. The clamp settles it. It cuts the free net's NaN loss from 13 of 30 epochs to none and holds best accuracy at 56.9%, level with the free baseline's 56.8% and above the Dale's-law recipe: stability with no loss of the signed net's expressivity.

Methods

Across this program a recurrent spiking network is trained on the Spiking Heidelberg Digits (SHD) [**1**], a 20-class spoken-digit benchmark. The **free** variant, whose recurrent weights are **signed** (a neuron may both excite and inhibit, breaking Dale's law), diverges during training as its loss becomes not-a-number (NaN). A finer integration step (exp061) and weight decay (exp063) both failed to stabilise this free signed net, and only enforcing Dale's law (exp062) worked, at an accuracy cost. Reading the forward pass explains why: the conductance-based leaky integrate-and-fire (LIF) neuron relaxes each timestep toward a steady-state voltage that integrates in closed form to

$$v_\infty = (g_L E_L + g_e E_e + g_i E_i)/g_{\text{tot}}, \quad g_{\text{tot}} = g_L + g_e + g_i,$$

where

- v_∞ is the steady-state (asymptotic) membrane voltage the neuron is driven toward;
- g_L, g_e, g_i are the leak, excitatory, and inhibitory conductances;
- E_L, E_e, E_i are their respective reversal (equilibrium) potentials;
- g_{tot} is the total conductance, the sum of the three.

With **signed** weights an excitatory or inhibitory conductance can itself go negative, so g_{tot} can cross zero and v_∞ diverges to NaN. That is why exp060 saw a NaN as early as epoch 2 even with tiny weights: the cause is the **sign** of the weights, not their magnitude.

The reserved fix is a **forward-pass state clamp**: during the forward pass, floor each conductance at 0 on every timestep (a conductance cannot be physically negative), which keeps $g_{\text{tot}} \geq g_L > 0$ and bounds v_∞ between the reversal potentials, while the **weights stay signed and expressive**. Three cells test the clamp at $\Delta t = 1$ ms, single seed, full scale:

1. **free (baseline)**: the signed net with no Dale's-law constraint, the one that diverges;
2. **free + state clamp**: the same net with the forward-pass state clamp added. Does the clamp remove the NaN divergence?
3. **free + clamp + decay 0.1**: exp063′s strongest weight decay reached 63.4% but could not stabilise the net on its own. With the clamp now doing the stabilising, does clamp plus decay top the program?

Parameter	Value
Dataset	SHD, 1000-sample subset
Model	COBANet (conductance-based network), 256 hidden units, all four recurrent weight blocks trainable, signed (no Dale's law)
Cells	free · free + clamp · free + clamp + weight-decay 0.1
State clamp	floor conductances at 0, cap magnitude at 100 µS each timestep
Integration	$\Delta t = 1$ ms, $T = 1000$ ms
Seeds	1 (seed 42)
Training	30 epochs, learning rate 0.001, batch size 32, surrogate-gradient backpropagation through time (BPTT) [**2**]
Compute	runpod

Success. The clamp trains the free net NaN-free with bounded dynamics, at accuracy at least matching Dale's law: stability that keeps the signed net's expressivity. **Kill:** if the clamp leaves any NaN present, the $g_{\text{tot}} \leq 0$ mechanism is not the whole cause.

Results

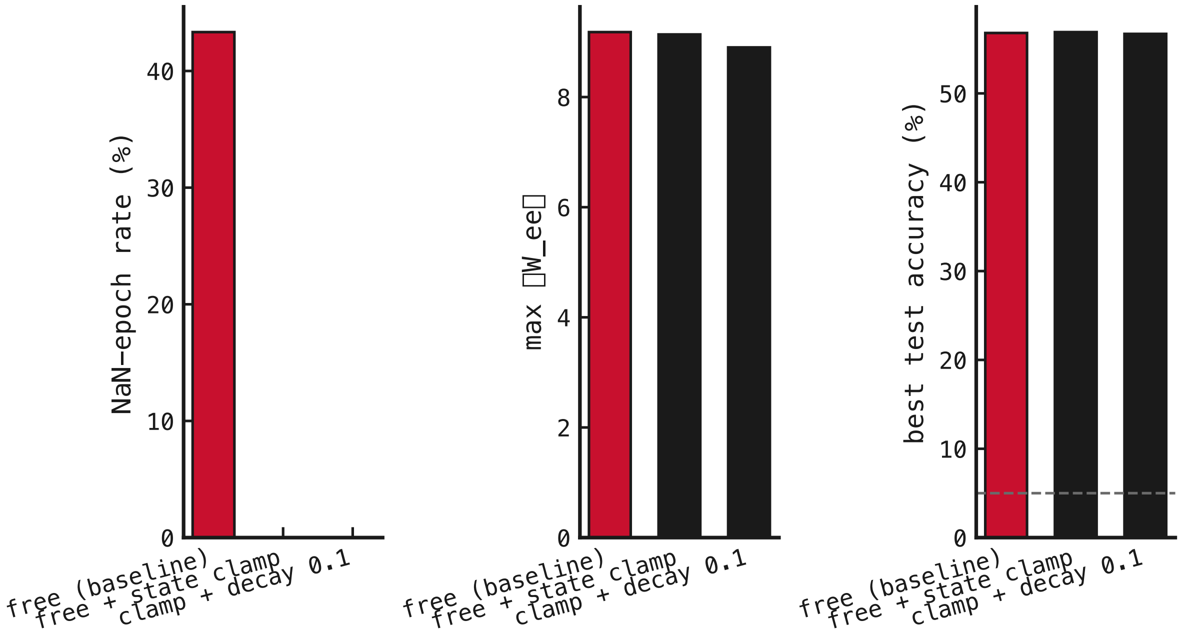

Figure 1: The three recipes compared on three measures (free baseline in red, both clamped cells in black). Left panel: the percentage of the 30 training epochs whose loss became not-a-number (NaN). Middle panel: the largest recurrent excitatory-to-excitatory weight norm ‖W_ee‖ reached during training. Right panel: best test accuracy (percent), with the dashed line marking the 5% chance level. **What we see confirms the mechanism.** The free baseline (red) carries 13 of 30 epochs with a not-a-number (NaN) loss; both clamped cells (black) drop that to **zero**. Critically, the recurrent excitatory-to-excitatory weight norm ‖W_ee‖ is unchanged by the clamp (9.1 versus the free net's 9.2): the clamp bounds the conductance, not the weight, exactly as the total-conductance $g_{\text{tot}} \leq 0$ mechanism predicts. And accuracy is untouched: the clamped net reaches 56.9%, matching the free baseline's 56.8%.

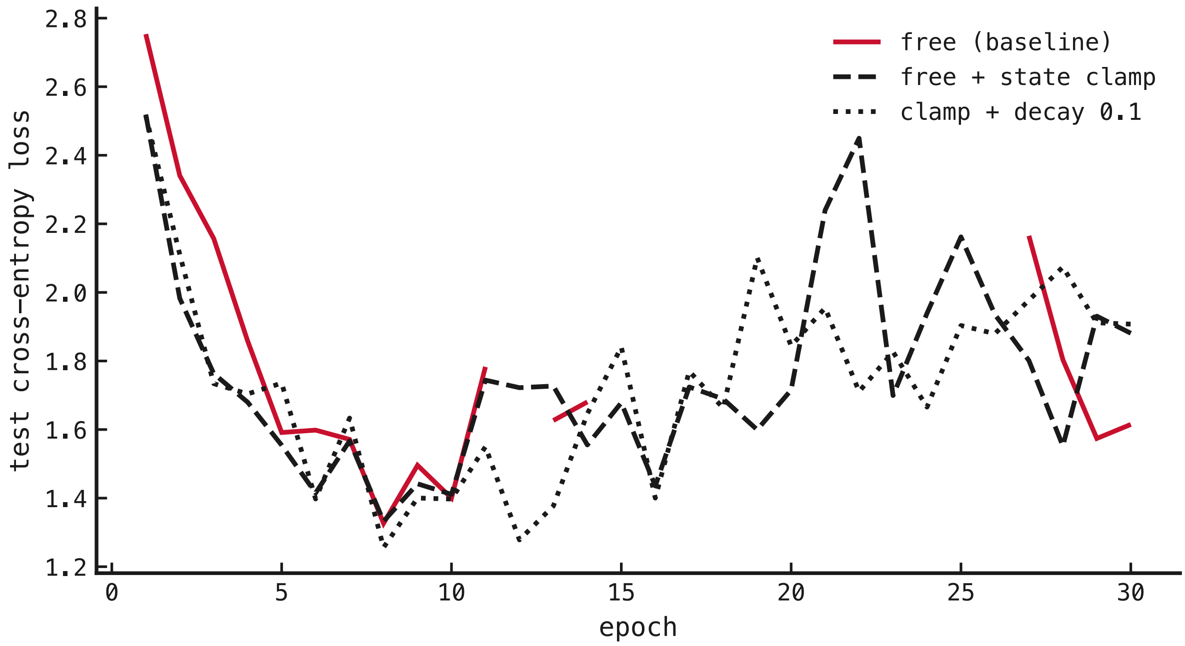

Figure 2: Test cross-entropy loss per epoch for the three recipes (x-axis: training epoch; y-axis: loss on the held-out test set). The free baseline (red) leaves gaps at the epochs where its loss diverges to not-a-number; the two clamped cells (near-black, distinguished by line style) run continuously from the first epoch to the last, so the clamp removes the divergence.

The clamp stabilises the signed net at no accuracy cost. The free baseline diverges as always: 13 of 30 epochs reach a not-a-number (NaN) loss, with a peak gradient norm (before clipping) of $\approx 10^4$. Add only the forward-pass state clamp and the divergence vanishes: 0 NaN epochs, the gradient norm bounded at ≈ 190, and best accuracy 56.9%, indistinguishable from the free net's 56.8%. The clamp is nearly transparent when the net behaves and bites only when a conductance would cross into the divergent $g_{\text{tot}} \leq 0$ regime.

The mechanism is confirmed, not merely patched. Two signatures pin it down. First, the weight norm ‖W_ee‖ is unchanged by the clamp (9.1 clamped versus 9.2 free): the weights grow exactly as before, so the clamp bounds the **state** (the conductance), not the weight. That is why weight decay, which bounds the weights, could not fix the divergence while the clamp can. Second, the effect is total (13 $\rightarrow$ 0 NaN epochs), as a hard floor on a divergent quantity should be, not the partial trend a soft regulariser gives.

The program's answer, ranked. Three recipes now train the net stably: Dale's law, the state clamp, and clamp plus weight decay. The state clamp is the best of them: it keeps the signed net's full accuracy (56.9%, clearing the Dale's-law recipe by several points) while removing the divergence, whereas Dale's law bought stability at the price of accuracy. The plan's registered goal was a stable recipe; the sharper answer is that the free net never needed **constraining**, only its **state bounding**.

What is left. Strong decay on top of the clamp did not lift accuracy here (56.7%), so exp063′s unstable 63.4% peak was a property of the diverging trajectory, not headroom the clamp preserves: the stable ceiling for this recipe sits near 56.9%. Confirming the result across random seeds, and testing whether the clamp shifts the network's γ (gamma) rhythm or its firing sparsity, are the natural follow-ups.

Caveat. Single seed. The stability effect is categorical (13 versus 0 NaN epochs) and so is safe to report, but the accuracy gaps between the stable recipes fall within a few points and want a three-seed confirmation before they can be ranked.

References

1. Cramer, Stradmann, Schemmel & Zenke — *The Heidelberg Spiking Data Sets for the Systematic Evaluation of Spiking Neural Networks*. 2022. doi:10.1109/TNNLS.2020.3044364
2. Neftci, Mostafa & Zenke — *Surrogate Gradient Learning in Spiking Neural Networks*. 2019. doi:10.1109/MSP.2019.2931595